IVT-GSCNet: Structure-Consistent and Semantically Aligned Unpaired Infrared-to-Visible Translation for UAV Remote Sensing Imagery

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***Abstract*—Uncrewed aerial vehicle (UAV) infrared (IR) remote sensing is pivotal for nighttime perception. However, IR images inherently suffer from low resolution and a lack of chromatic information, which restricts semantic scene understanding. To address the triad of challenges—noise contamination, semantic misalignment, and structural distortion—in unpaired IR-to-visible (IR2Vis) translation, this article proposes an IR2Vis translation gated spatial-channel network (IVT-GSCNet), a novel framework tailored for UAV remote sensing. Specifically, a gated feature refinement module (GFRM) is introduced to adaptively suppress sensor noise and background clutter within U-Net skip connections, ensuring clean feature propagation for reconstruction. Furthermore, to achieve precise spatial alignment without paired data, we propose a layer-weighted contrastive learning mechanism. By assigning higher weights to deep layers that encode global semantic features, our method significantly improves the preservation of overall scene structure and the localization of small objects in UAV imagery. Finally, a composite structural constraint—integrating edge, structural similarity, and smoothness losses—is implemented to safeguard geometric fidelity. Extensive experiments across four diverse multimodal datasets (including primary UAV-captured sets, MSRS, and M3FD) demonstrate that IVT-GSCNet significantly outperforms state-of-the-art (SOTA) methods in multidimensional perceptual quality [e.g., learned perceptual image patch similarity (LPIPS), kernel inception distance (KID), and blind/referenceless image spatial quality evaluator (BRISQUE)] and downstream semantic segmentation. Systematic ablation studies and parameter analyses further validate the individual module contributions and confirm our model's unique capability in balancing structural coherence and photorealism, establishing its practical value for low-illumination UAV perception.**

***Index Terms*—Frequency-spatial synergy, infrared (IR)-to-visible translation, saliency protection, semantic segmentation, uncrewed aerial vehicle (UAV) remote sensing, unpaired learning.**

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I. Introduction

UNCREWED aerial vehicles (UAVs) have become essential in remote sensing, widely used in precision agriculture [1], power line inspection [2], traffic monitoring [3], disaster search and rescue [4], military reconnaissance [5], and urban planning [6]. However, under long-distance high-altitude imaging or harsh conditions like nighttime and heavy haze, UAV-captured visible (RGB) images often suffer from significant quality degradation [7], [8], [9]. To alleviate the informational limitations of standard three-channel visible data, spectral super-resolution techniques such as coarse-to-fine reconstruction networks [10] and wavelet-based hybrid asymmetric architectures like WHANet [11] have been developed to recover high-dimensional spectral signatures. Alternatively, infrared (IR) sensors provide all-weather target contours via thermal radiation [12], [13], yet lack color information. Thus, IR-to-visible (IR2Vis) translation has emerged as a critical technique to enhance scene understanding [14], [15], [16]. Early translation approaches relied on paired data. Pix2pix [17] pioneered cGANs, and perceptual loss [18] improved visual realism. Architectural backbones evolved from U-Net [19], Res-UNet [20], Bridge-U-Net [21], to UNet++ [22], with additional refinements like dense connections [23], SE modules [24], and depthwise convolutions [25]. Despite these paired UAV data is scarce, and traditional skip connections often fail to suppress IR-specific noise [26]. Unsupervised paradigms emerged to mitigate data scarcity. CycleGAN [27] utilized cycle-consistency, but often faces geometric collapse. MUNIT [28] decouples style for multimodal tasks, yet struggles with structural fidelity. GcGAN [29] and neural optimal transport [30] model nonlinear mappings, while recent contrastive learning approaches—including CUT [31] and DCLGAN [32]—enhance feature alignment. Advanced frameworks such as QS-Attn [33] and UNSB [34] further introduce semantic constraints for robust domain bridging. Among them, UNSB is a method based on a neural Schrödinger bridge.

Recent trends have shifted toward transformer-based architectures [35], [36], [37] and fusion strategies [38], [39]. These models, along with lightweight variants [40], [41] aim to capture long-range dependencies for restoration [42] and registration [43]. Furthermore, advanced diffusion-based

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translation models, such as the joint cycle-consistent framework CycleDiff [44], along with other generative approaches [45] and attention-aware fusion [46], [47], have redefined structural preservation. However, these complex models face deployment bottlenecks in real-time UAV surveillance due to computational overhead [41].

Despite these efforts, translating natural image-oriented methods to UAV data remains difficult due to extreme small-target proportions and large modality gaps [48], [49]. Although specialized airborne translation methods, such as the diverse and realistic DR-AVIT framework [50], have been proposed to align aerial modalities, most conventional methods often lead to the "alignment-realism dilemma," compromising either layout or texture. While recent studies leverage blind/referenceless image spatial quality evaluator (BRISQUE) [51], MMD [52], or task-specific metrics [53] as optimization objectives to guide training, they lack explicit noise-semantic decoupling, resulting in hallucinations [54]. Motivated by these limitations, we propose the IVT-GSCNet, integrating gated feature purification [55] and layer-weighted contrastive strategies to bridge the gap between thermal signals and high-level semantic understanding.

Although the aforementioned methods have achieved significant progress in general image translation and certain IR translation tasks, several crucial issues remain unresolved in the complex UAV-based IR2Vis task.

Issue 1: Large cross-modal differences make it challenging for conventional single-network architectures to simultaneously preserve global semantic consistency and local texture reconstruction under unpaired training conditions, often resulting in structural distortions or semantic misalignment.

Issue 2: High-altitude imaging and IR sensor noise introduce significant interference, making it difficult for conventional networks to distinguish targets from the background. Small target signals are easily overwhelmed, adversely affecting image clarity and target recognizability.

Issue 3: In UAV images, small targets occupy a low proportion and IR-visible signals differ substantially in modality, making it difficult for conventional unpaired methods to achieve spatial-semantic alignment and geometric consistency. This often leads to target misplacement or structural distortion in the generated images.

To address the above challenges, this article proposes IVT-GSCNet, an end-to-end network that combines a holistic encoder–decoder design with noise suppression modules and layer-weighted contrastive learning mechanisms to achieve high-fidelity UAV IR2Vis translation.

The main contributions of this article are as follows.

1) This article proposes an end-to-end UAV IR2Vis translation network, namely, IR2Vis translation gated spatial-channel network (IVT-GSCNet), which employs a holistic encoder–decoder architecture to robustly process high-noise inputs while simultaneously preserving global semantic consistency and local details, effectively addressing modality differences and semantic consistency issues.

2) This article proposes a gated feature refinement module (GFRM), which introduces a spatially adaptive channel attention mechanism in the skip connections to suppress noise and enhance small target features, providing a clean feature flow for image reconstruction and solving the problem of small target loss in complex backgrounds.

3) This article proposes a layer-weighted contrastive learning and composite structure constraint mechanism, which leverages multilayer weighted PatchNCE combined with edge, structural similarity, and total variation (TV) constraints to achieve spatial-semantic alignment and geometric preservation, ensuring small target positioning and structural integrity while generating natural and realistic visible images.

The pipeline of the remaining sections of this article is as follows. In Section II, we detail the architecture of the network. In Section III, we present this work, including the experimental results and relevant comparisons to validate the effectiveness of our approach. Finally, in Section IV, we summarize this work.

II. Methodology

A. Overview

The overall framework of the proposed IVT-GSCNet is illustrated in Fig. 1. Given a pair of unpaired IR image $\text{Infrared}_{\text{Real}}$ from the source domain and visible image $\text{Visible}_{\text{Real}}$ from the target domain, our goal is to learn a mapping function that synthesizes a visible image $\text{Visible}_{\text{Fake}}$ with realistic optical textures while maintaining consistent geometric structures with the input IR image. The proposed architecture is built upon three collaborative pillars designed to address the unique challenges of UAV-based IR translation: 1) a gated generator (G) optimized for hierarchical feature purification and noise suppression; 2) a patch projection network (F) facilitating layer-weighted multiscale contrastive alignment; and 3) a PatchGAN discriminator (D) for adversarial distribution matching. In the proposed pipeline, the input IR image is first processed by the gated encoder to extract multilayer features. Specifically, GFRM acts as adaptive filters before each skip connection to mitigate sensor-induced noise propagation (Issue 1). Simultaneously, to ensure robust semantic alignment without paired data (Issue 2), we extract hierarchical features and project them into a shared embedding space via network F, where a layer-weighted contrastive strategy is employed to anchor domain-invariant semantics. Finally, to prevent structural hallucinations caused by the modality gap (Issue 3), an explicit structural constraint branch integrating edge gradients, structural similarity, and TV is implemented to safeguard the geometric skeleton. The entire framework is jointly optimized through a composite objective function, achieving a state-of-the-art (SOTA) balance between perceptual photorealism and structural integrity.

B. Gated Generator With Hierarchical Refinement and Output Projection (OP)

As illustrated in Fig. 2, the proposed gated generator G is architected as a symmetric encoder–decoder framework, meticulously optimized for hierarchical feature purification and cross-modal texture synthesis. The network is composed of four primary synergistic components: the content encoder (CE), the style decoder (SD), GFRM, and the OP module.

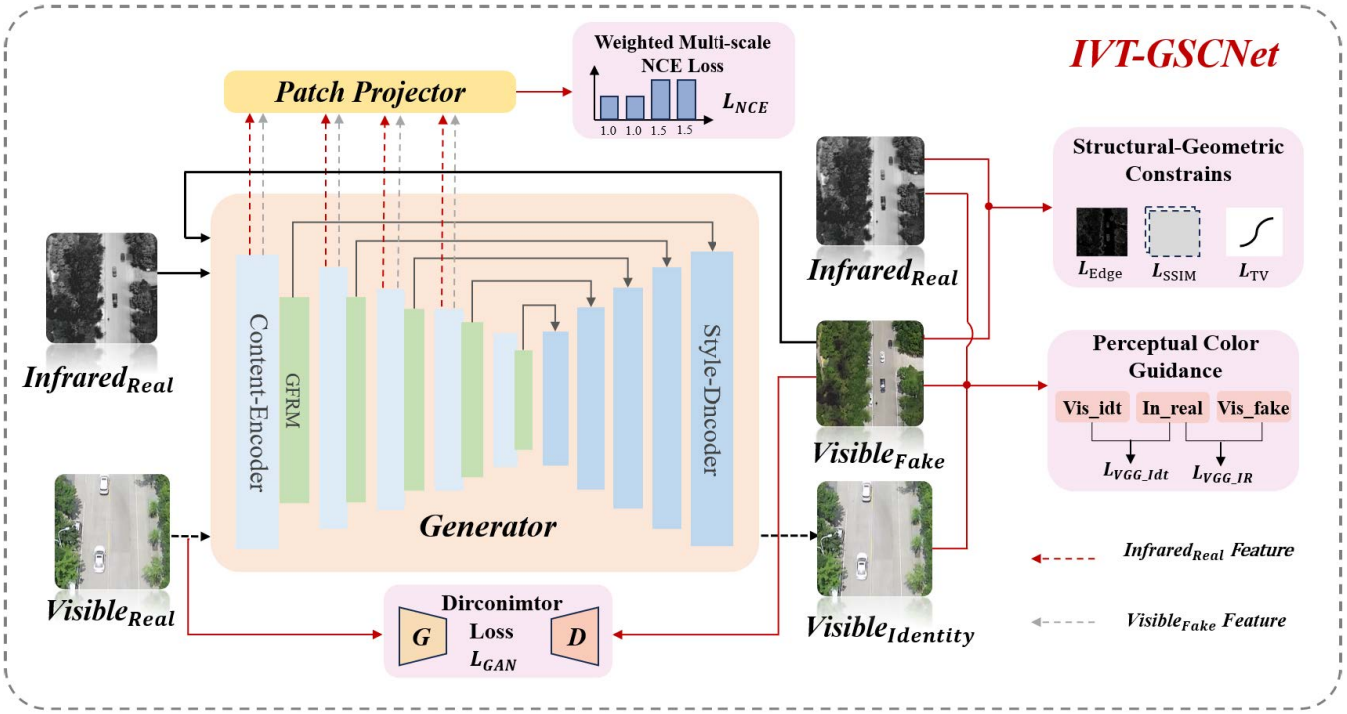


Fig. 1. Overall architecture of the proposed IVT-GSCNet. The framework consists of a gated generator (G), a patch projector (F), and a discriminator (D).

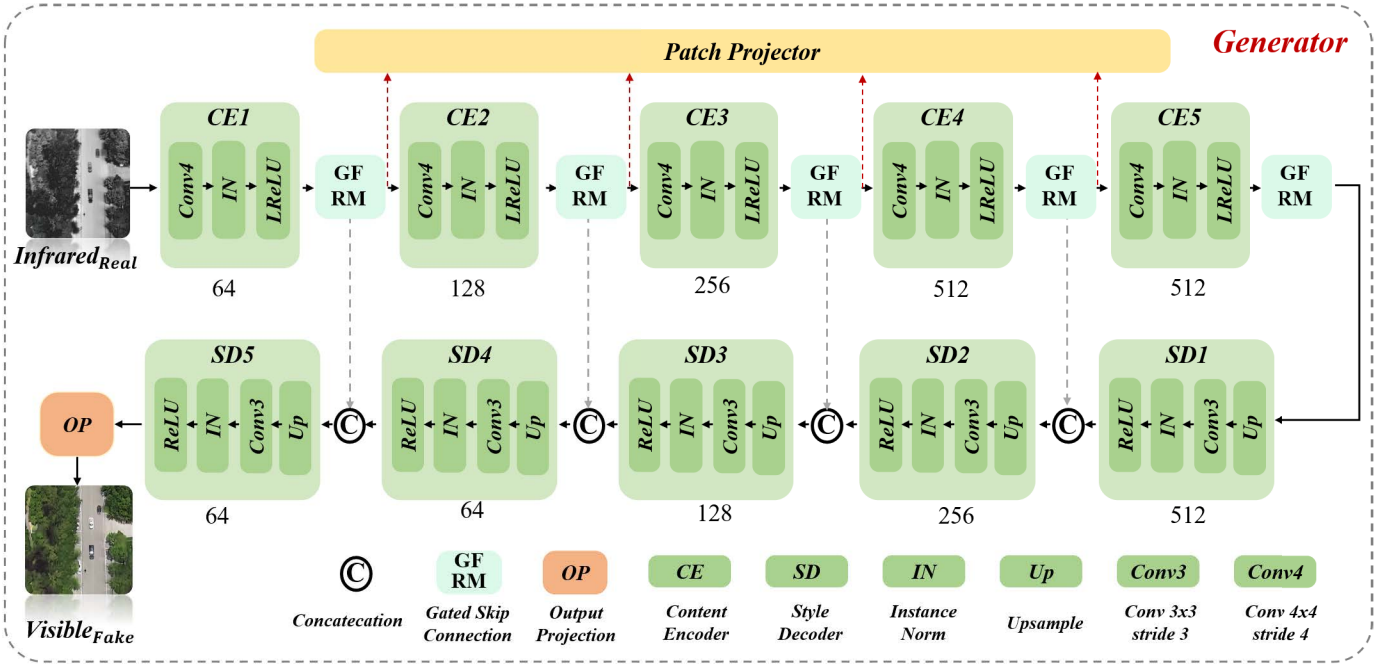


Fig. 2. Detailed architecture of the proposed gated generator. The framework follows a symmetric encoder–decoder design comprising the CE and SD.

1) Hierarchical CE: The encoder part comprises five hierarchical blocks, denoted as $CE1$ to $CE5$ in Fig. 2. Each block is designed to progressively compress spatial resolution while expanding the channel depth from 64 to 512. Following the data flow depicted in the diagram, each CE block utilizes a 4×4 convolutional layer with a stride of 4, followed by instance normalization (IN) and leaky ReLU activation. This aggressive yet controlled downsampling strategy allows the network to extract robust hierarchical representations, capturing both local edge gradients and high-level semantic manifolds. To prevent the direct propagation of IR sensor noise, the features extracted by each CE block are not fed directly into the decoder but are first processed by the GFRM to ensure feature purity.

2) Hierarchical SD: The decoder subnetwork, consisting of blocks $SD1$ to $SD5$, is responsible for reconstructing photo-realistic visible-light textures. As shown in the bottom branch of Fig. 2, each SD block mirrors the encoder's hierarchy to

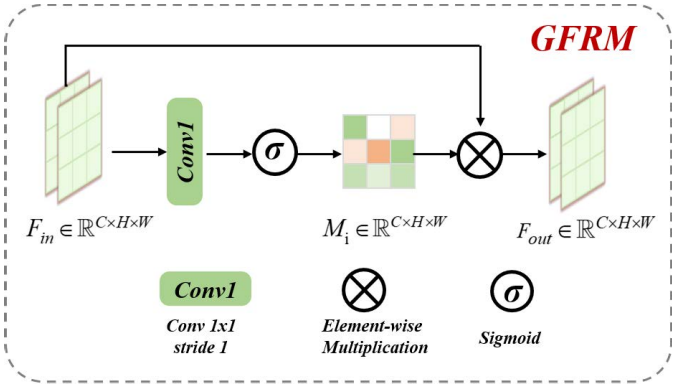


Fig. 3. Internal schematic of the GFRM.

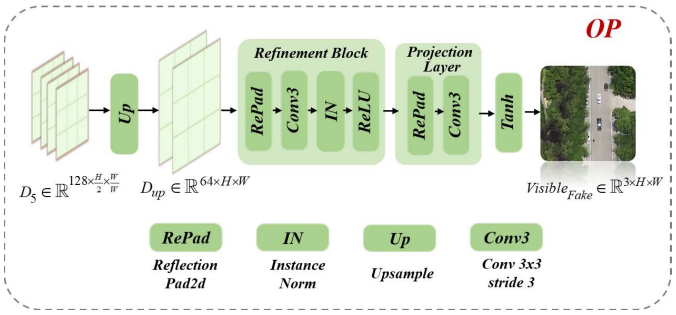


Fig. 4. Architecture of the OP module.

maintain structural symmetry. The decoding process starts with a bilinear upsampling layer, followed by a 3×3 convolution with a stride of 3, IN, and ReLU activation. At each stage, the decoder integrates purified features from the gated skip connections via a concatenation operation. This hierarchical fusion mechanism ensures that the stylistic reconstruction is consistently anchored by noise-free geometric skeletons provided by the CE.

3) Gated Feature Refinement Module: The GFRM is designed not as a common attention layer, but as a context-aware signal regulator specifically tailored for the low signal-to-noise ratio (SNR) of IR imagery. In standard skip connections, the direct concatenation of encoder features often introduces persistent thermal artifacts into the decoder, hindering high-frequency texture synthesis. To decouple target-relevant features from such contamination, the GFRM employs a point-wise gating mechanism. By modulating the spatial-channel activations without downsampling, it effectively acts as a feature purifier, ensuring that only the refined geometric skeletons—essential for UAV target localization—are propagated through the network hierarchy.

Compared to conventional gating or attention mechanisms, such as coordinate attention (CA) [56] and spatial gating units (SGUs) [57], the proposed GFRM is designed specifically to address the unique challenges of UAV IR imagery. Traditional mechanisms often employ global pooling or spatial downsampling to squeeze feature maps, which inevitably discards fine-grained spatial details. However, UAV IR images typically suffer from low SNRs, small target footprints, and heavy background clutter, where noise and weak target signals are spatially intertwined. Aggressive spatial squeezing would blur tiny targets and mix them with background noise. To circumvent this, the GFRM avoids any downsampling and employs a lightweight 1×1 convolution followed by a Sigmoid activation. This structure facilitates joint spatial-channel modulation, enabling the network to filter out high-frequency IR noise at a pixel-wise level while preserving the structural integrity of weak targets. Furthermore, this simple yet effective design incurs negligible computational overhead, making it highly suitable for real-time deployment on resource-constrained UAV hardware platforms.

As detailed in the internal schematic in Fig. 3, the module takes the raw encoder feature $F_{in} \in \mathbb{R}^{C\times H\times W}$ as input. A spatially adaptive gating mask M_i is computed by passing the feature through a 1×1 convolutional layer and a Sigmoid function (σ)

$$M_i = \sigma\left(\text{conv}_{1\times 1}\left(F_{in}\right)\right). \tag{1}$$

Subsequently, the refined output F_{out} is synthesized through ($\odot$)

$$F_{out} = F_{in} \odot M_i \tag{2}$$

where $M_i \in \mathbb{R}^{C\times H\times W}$ denotes the content-aware 3-D attention weights, $\sigma(\cdot)$ represents the Sigmoid activation function that squashes values into the range $[0, 1]$, and $\odot$ signifies the element-wise multiplication (Hadamard product). F_{in} and F_{out} correspond to the feature representations before and after the purification process, respectively.

4) OP Module: To facilitate a smooth transition from deep latent features to the RGB pixel space, we design the OP module as the final stage of the generator. As illustrated in the detailed pipeline in Fig. 4, the final 128-channel decoder feature map $D_5 \in \mathbb{R}^{128\times H/2\times W/2}$ is first upsampled via bilinear interpolation to match the target resolution $H \times W$ with a reduced channel depth of 64

$$D_{up} = \text{Upsample}_{bilinear}\left(D_5\right). \tag{3}$$

The upsampled features then pass through a refinement block utilizing reflection padding (RefPad), IN, and ReLU to enhance local detail consistency

$$D_{refine1} = \text{ReLU}\left(\text{IN}\left(\text{Conv}_{3\times 3}\left(\text{Pad}_{refl}\left(D_{up}\right)\right)\right)\right) \tag{4}$$

$$D_{refine2} = \text{ReLU}\left(\text{IN}\left(\text{Conv}_{3\times 3}\left(\text{Pad}_{refl}\left(D_{refine1}\right)\right)\right)\right). \tag{5}$$

Finally, a projection layer with Tanh activation synthesizes the final image

$$I_{out} = \text{Tanh}\left(\text{Conv}_{3\times 3}\left(D_{refine2}\right)\right) \tag{6}$$

where D_{up} is the upsampled feature map with spatial resolution $H \times W$, while $D_{refine1}$ and $D_{refine2}$ denote the intermediate refined states of the feature flow. $\text{Pad}_{refl}(\cdot)$ represents the RefPad operation used to suppress boundary artifacts, $\text{IN}(\cdot)$ is the IN, and $\text{Tanh}(\cdot)$ maps the output intensity to the range $[-1, 1]$ to match the target RGB distribution.

C. Cross-Modal Semantic Correspondence via Layer-Weighted Multiscale Contrastive Learning

In the absence of paired training data, ensuring robust semantic correspondence between the input IR domain and

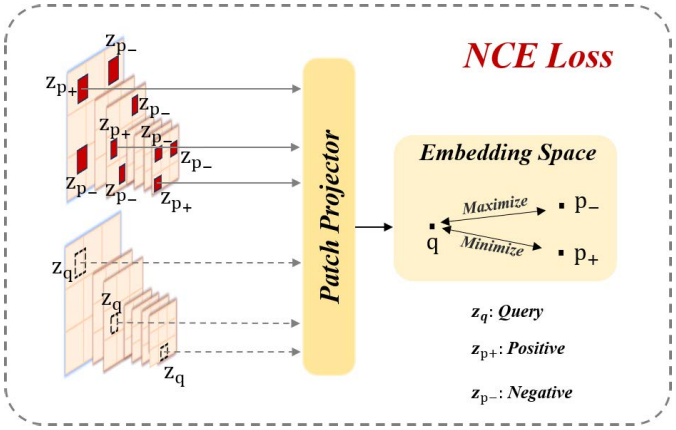


Fig. 5. Schematic of the layer-weighted multiscale PatchNCE mechanism.

synthesized visible domain is a formidable challenge. To address this, we propose a layer-weighted multiscale PatchNCE mechanism, as illustrated in Fig. 5. Unlike standard contrastive methods that treat all hierarchical layers equally, our approach assigns layer-specific importance to effectively navigate the alignment-realism dilemma.

1) Feature Extraction and Projection: Following the patch-wise contrastive learning framework, we extract intermediate feature maps from the first four stages of the CE ($CE1$ to $CE4$). These hierarchical representations, after being purified by the GFRM, are fed into the patch projector, which comprises two-layer multilayer perceptron heads H_l. The projector maps the high-dimensional features into a shared 256-D embedding space. To ensure stable optimization and robust similarity measurement, the projected embeddings z_l are L_2-normalized to reside on a unit hypersphere

$$z_l = \frac{H_l\left(F_l^{refined}\right)}{\left\|H_l\left(F_l^{refined}\right)\right\|_2} \tag{7}$$

where $F_l^{refined}$ represents the refined feature at layer l, and $\|\cdot\|_2$ denotes the L_2 norm.

2) Layer-Weighted Contrastive Formulation: For each synthesized image $\hat{I}_{vis}$, we randomly sample $Q = 256$ query patches across the selected layers. As visualized in the *embedding space* of Fig. 5, for a query z_q at a specific spatial location in layer l, its positive sample z_{p+} is the patch at the identical coordinate in the input IR image, while $N = 512$ negative samples z_{p-} are randomly drawn from other spatial positions within the same IR feature map.

The core objective is to pull the query toward its positive counterpart (minimize distance) while pushing it away from all negatives (maximize distance) in the embedding manifold. The multilayer weighted PatchNCE loss is formulated as

$$\mathcal{L}_{\text{NCE}} = \sum_{l=1}^{L} w_l \cdot \mathbb{E} \times \left[-\log\left(\frac{\exp\left(z_q^\top z_{p+}/\tau\right)}{\exp\left(z_q^\top z_{p+}/\tau\right) + \sum_{n=1}^{N}\exp\left(z_q^\top z_{p-}^{(n)}/\tau\right)}\right)\right] \tag{8}$$

where $L = 4$, w_l represents the layer-specific weight coefficients, and τ denotes the temperature scaling factor. Following the standard configurations in self-supervised contrastive learning [58], [59] and patch-wise image-to-image translation [31], the temperature scaling factor τ is directly set to 0.07. Theoretically, τ plays a pivotal role in regulating the penalty strength on hard negative samples. A smaller τ forces the model to scale up gradients for challenging negative pairs, which helps the network learn highly discriminative features; however, an excessively small temperature can lead to gradient instability or overfitting to noisy negatives. Conversely, a larger τ tends to treat negative samples too uniformly, diminishing the contrastive strength. Thus, this widely accepted value of 0.07 is employed here to ensure stable training convergence and optimal feature representation without introducing additional hyperparameter search. In our implementation, we adopt a strategic weight allocation of $w_l = [1.0, 1.0, 1.5, 1.5]$. This configuration is specifically designed to emphasize deep semantic anchoring in the final two stages, ensuring global layout consistency while granting the shallow layers sufficient generative flexibility for realistic texture synthesis. A detailed sensitivity analysis justifying this specific parameter selection is provided in Section III-E.

D. Geometry-Guided Structural Constraints

IR images inherently preserve reliable geometric information, such as object boundaries and scene layouts, which are crucial for consistent drone-based perception. However, during cross-modal translation, these structures are prone to distortion or loss due to the vast modality gap. To mitigate this, we introduce a composite structural constraint branch that operates at three complementary levels: pixel-level edge preservation, regional structural coherence, and spatial smoothness.

1) Edge Consistency via Sobel Operator: Object boundaries in IR images are robust to illumination variations and serve as the *geometric skeleton* for the translation process. We utilize the Sobel operator to extract gradient magnitude maps from both the input IR image I_{ir} and the synthesized visible image $\hat{I}_{vis}$. The horizontal (K_x) and vertical (K_y) Sobel kernels are defined as

$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad K_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}. \tag{9}$$

The gradient components for the IR and synthesized visible domains are computed via convolution

$$G_x^{ir} = K_x * I_{ir}, \quad G_y^{ir} = K_y * I_{ir} \tag{10}$$

$$G_x^{vis} = K_x * \hat{I}_{vis}, \quad G_y^{vis} = K_y * \hat{I}_{vis}. \tag{11}$$

The total gradient magnitude maps are then derived as follows:

$$G^{ir} = \sqrt{\left(G_x^{ir}\right)^2 + \left(G_y^{ir}\right)^2} \tag{12}$$

$$G^{vis} = \sqrt{\left(G_x^{vis}\right)^2 + \left(G_y^{vis}\right)^2}. \tag{13}$$

To ensure scale invariance and prioritize structural alignment over absolute intensity, we minimize the L_1 distance between the normalized magnitude maps

$$\mathcal{L}_{Edge} = \mathbb{E}\left[\left\| \frac{G^{ir}}{\max\left(G^{ir}\right)} - \frac{G^{vis}}{\max\left(G^{vis}\right)} \right\|_1\right] \tag{14}$$

where $\mathcal{L}_{Edge}$ enforces strict boundary alignment, effectively preventing the target misplacement often seen in baseline methods.

2) Structural Similarity and TV: While edge loss preserves sharp contours, it does not explicitly capture higher-order structural relationships such as region homogeneity and contrast consistency. To address this, we employ the structural similarity index (SSIM) loss. For two corresponding local patches x and y, the mean (μ), variance (σ^2), and covariance (σ_{xy}) are formulated as

$$\mu_x = \frac{1}{N}\sum_{i=1}^{N} x_i, \quad \sigma_x^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu_x)^2 \tag{15}$$

$$\sigma_{xy} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu_x)\left(y_i - \mu_y\right). \tag{16}$$

The SSIM index is then computed using luminance, contrast, and structural comparisons

$$SSIM(x,y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{17}$$

where C_1 and C_2 are small constants to avoid division by zero. The resulting SSIM loss $\mathcal{L}_{SSIM}$ is defined as

$$\mathcal{L}_{SSIM} = 1 - \mathbb{E}\left[SSIM\left(I_{ir}(p), \hat{I}_{vis}(p)\right)\right]. \tag{18}$$

By combining $\mathcal{L}_{Edge}$ and $\mathcal{L}_{SSIM}$, the model maintains both fine-grained geometric skeletons and regional structural integrity.

Furthermore, to mitigate checkerboard artifacts and high-frequency noise that commonly plague GAN-based translations, we implement TV regularization

$$\mathcal{L}_{\mathrm{TV}} = \frac{1}{C}\sum_{c=1}^{C}\sum_{h=1}^{H-1}\sum_{w=1}^{W-1} \sqrt{\left(\hat{I}_{h+1,w,c} - \hat{I}_{h,w,c}\right)^2 + \left(\hat{I}_{h,w+1,c} - \hat{I}_{h,w,c}\right)^2}. \tag{19}$$

By penalizing large intensity differences between neighboring pixels, $\mathcal{L}_{TV}$ promotes spatial smoothness while preserving essential structural boundaries, ensuring the generated images are perceptually natural and realistic.

E. Full Objective Function

The final optimization objective of our proposed IVT-GSCNet is formulated as a weighted combination of adversarial loss, contrastive loss, and structural-perceptual constraints. This synergistic multiloss framework enables the model to simultaneously optimize for photorealistic appearance, semantic alignment, and geometric integrity. The overall objective function is defined as

$$\begin{aligned}\mathcal{L}_{\mathrm{total}} = {} & \lambda_{\mathrm{GAN}}\mathcal{L}_{\mathrm{GAN}} + \lambda_{\mathrm{NCE}}\mathcal{L}_{\mathrm{NCE}} + \lambda_{\mathrm{Edge}}\mathcal{L}_{\mathrm{Edge}} \\ & + \lambda_{\mathrm{SSIM}}\mathcal{L}_{\mathrm{SSIM}} + \lambda_{\mathrm{TV}}\mathcal{L}_{\mathrm{TV}} + \lambda_{\mathrm{VGG}}\mathcal{L}_{\mathrm{VGG}} \\ & + \lambda_{\mathrm{Idt}}\mathcal{L}_{\mathrm{Idt}}\end{aligned} \tag{20}$$

where the hyperparameters λ control the relative importance of each objective. Specifically, adversarial loss ($\mathcal{L}_{GAN}$) employs a 70 $\times$ 70 PatchGAN discriminator to ensure the synthesized images approximate the statistical distribution of real visible imagery. Contrastive loss ($\mathcal{L}_{NCE}$), implemented with our hierarchical weighting scheme, anchors semantic correspondence without paired data. To bridge the modality gap in textures and colors, we incorporate perceptual loss ($\mathcal{L}_{VGG}$), which computes the L_1 distance between high-level feature maps extracted from a pretrained VGG-19 network, guiding the generator toward a natural color manifold. Furthermore, the identity loss ($\mathcal{L}_{Idt}$) is applied when visible images are used as input to maintain color fidelity and content consistency. Collectively, while $\mathcal{L}_{GAN}$ and $\mathcal{L}_{VGG}$ drive the stylistic photorealism, the triad of structural terms ($\mathcal{L}_{Edge}$, $\mathcal{L}_{SSIM}$, and $\mathcal{L}_{TV}$) serves as the *geometric stabilizer*, ensuring that the generated images are not only visually authentic but also strictly adhered to the physical layout of the IR source. Through this end-to-end joint optimization, IVT-GSCNet successfully mitigates noise propagation and structural distortion in UAV sensing environments.

III. Experiment and Analysis

A. Datasets

We evaluate the proposed IVT-GSCNet on two tasks: IR2Vis image translation and the downstream semantic segmentation task. For the translation task, we construct two primary unpaired training datasets from the AVIID dataset [60], and further evaluate the cross-domain generalizability on both the newly introduced M3FD dataset [61] and the MSRS dataset [62]. For the downstream semantic segmentation task, we use the MSRS dataset [62].

1) *Dataset A: Intraphase Cross-Modal Translation Dataset:* We merge all images (including both IR and visible modalities) from AVIID3-unpaired-5, then randomly split them into training and test sets at an 8:2 ratio. This yields 1024 unpaired IR-visible image pairs (1024 IR and 1024 visible) for training, and 256 IR images for testing (with corresponding visible images held out for reference).
2) *Dataset B: Cross-Temporal Cross-Modal Translation Dataset:* To further increase data diversity and evaluate the cross-domain generalization ability, we construct an additional training set by combining the nighttime IR images from AVIID2-unpaired-5 and the daytime visible images from AVIID1-unpaired-5. This results in 498 unpaired IR-visible image pairs (498 IR images from nighttime scenes and 498 visible images from daytime scenes), with no requirement for alignment between modalities.
3) *Dataset C: Multiscene Multimodal Translation Dataset:* To evaluate the cross-domain generalizability across diverse outdoor environments, we extract a multiscene subset from the detection split of the public M3FD

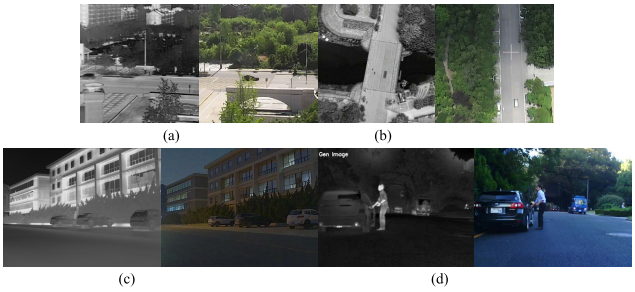


Fig. 6. Representative samples from the three utilized datasets. (a) Dataset A: cross-modal translation (mixed scenes from AVIID3-unpaired-5). (b) Dataset B: cross-modal translation (nighttime IR to daytime visible). (c) Dataset C: M3FD dataset with multiscene multimodal outdoor images. (d) Dataset D: MSRS dataset with semantic segmentation labels.

dataset [61]. This yields 1069 unpaired IR-visible image pairs (1069 IR, 1069 visible, totaling 2138 images) for training, and 119 IR images for testing (with corresponding 119 visible images held out for reference). During training, these co-registered images are shuffled to construct an unpaired setting, rigorously testing the model's translation fidelity and structural preservation capabilities under complex topographic layouts and varied lighting conditions.

4) *Dataset D: Downstream Semantic Segmentation and Translation Utility Dataset:* The MSRS dataset provides pixel-aligned IR-visible image pairs and comprehensive semantic annotations. In contrast to the unpaired sets, this dataset facilitates quantitative assessment of semantic consistency. Following standard protocols, we employ an intensity-based filtering strategy to exclude nighttime scenarios by removing images with a global pixel mean below 60. This refinement results in a daytime-focused subset comprising 536 IR-visible pairs for training and 361 pairs for testing, each with corresponding pixel-level labels. The adoption of this dataset is motivated by the need to verify whether IVT-GSCNet can effectively preserve discriminative semantic information sufficient for downstream high-level vision tasks. In addition, to thoroughly address the concern on dataset diversity, Dataset D is also utilized for evaluating the image-to-image translation task under aligned daytime conditions.

Fig. 6 visualizes representative samples from each dataset. Dataset A contains mixed scenes with moderate illumination variations, where IR and visible images capture the same scenes under diverse conditions. Dataset B consists of extreme cross-domain datasets, requiring the model to translate nighttime IR images to daytime visible images. Dataset C comprises complex multiscene outdoor environments featuring distinct buildings and road targets to evaluate generalizability. Dataset D provides aligned IR-visible pairs with pixel-level semantic annotations, enabling both aligned image translation and quantitative evaluation of downstream segmentation performance.

B. Implementation Details

The proposed IVT-GSCNet is implemented using the PyTorch framework and trained on an NVIDIA Quadro P5000 GPU with 16 GB of VRAM. All input images are uniformly resized to 256 $\times$ 256 pixels. For optimization, we employ the Adam optimizer with momentum parameters $\beta_1 = 0.5$ and $\beta_2 = 0.999$. The initial learning rate for both the generator and the discriminator is set to 0.0002. The model is trained for a total of 400 epochs with a batch size of 4. Data augmentation is performed via random horizontal flipping to enhance model generalization. To ensure a rigorous and fair evaluation, we compare our method against seven SOTA baselines. All baseline models are trained on our datasets using their official source codes and recommended hyperparameters. For the downstream semantic segmentation evaluation, we utilize a DeepLabV3 + [63] model pretrained on ImageNet and subsequently fine-tuned on the MSRS training set.

C. Evaluation Metrics

To comprehensively evaluate the translation quality and generalizability from multiple dimensions, we employ four distinct metrics encompassing both distribution-level and perceptual-level assessments as follows.

1) Kernel Inception Distance (KID) [52]: KID measures the MMD between high-level feature representations of real and synthesized visible images, extracted from a pretrained Inception-v3 network. It is highly robust to small sample sizes and serves as a reliable index of feature distribution alignment. Lower values indicate better translation quality.

2) Mutual Information (MI) [64]: MI quantifies the statistical dependence and information overlap between the input IR image and the synthesized visible image, thereby evaluating spatial-semantic and layout alignment. Higher values denote superior structural consistency.

3) BRISQUE [51]: BRISQUE is a reference-free image quality metric based on natural scene statistics. It extracts local normalized luminance coefficients to evaluate the naturalness and the presence of distortions (such as blurring or artifacts) in the synthesized images. Lower values indicate higher perceptual quality.

4) Learned Perceptual Image Patch Similarity (LPIPS) [65]: To expand the evaluation dimension, we introduce LPIPS. It measures the distance between deep feature maps of local image patches extracted from a pretrained VGG-16 network, providing a reliable measure of human-aligned perceptual similarity and high-frequency texture fidelity. Lower values represent higher similarity to real target domain images.

D. Comparison With SOTA Methods

To comprehensively evaluate the performance of IVT-GSCNet, we compare our proposed method with eight representative baseline models. CycleGAN [27] is a pioneering approach that utilizes cycle-consistency loss for content preservation. CUT [31] further improves by replacing cycle constraints with PatchNCE, reducing computational overhead while optimizing texture fidelity. MUNIT [28] disentangles content and style codes to enable multimodal modeling, supporting flexible domain transfer. QS [33] introduces a query-selected attention mechanism to enhance feature patch relevance during cross-domain mapping. GcGAN [29]

TABLE I
QUANTITATIVE COMPARISON ON DATASET A

Method	KID(↓)	MI(↑)	BRISQUE(↓)	LPIPS(↓)
CycleGAN	0.1231	0.9752	21.5338	0.6392
CUT	0.0629	0.9498	20.1492	0.6389
UNSB	0.0688	0.9875	19.5590	0.6492
QS	0.0964	1.0586	19.6954	0.6602
MUNIT	0.1460	0.5034	21.1717	0.6613
NOT	0.1473	1.0000	28.1098	0.6865
GcGAN	0.1185	0.7936	19.0351	0.6617
CycleDiff	0.1137	1.0696	19.9379	0.7085
Ours	**0.0542**	**1.3660**	**18.3021**	**0.6308**

TABLE II
QUANTITATIVE COMPARISON ON DATASET B

Method	KID(↓)	MI(↑)	BRISQUE(↓)	LPIPS(↓)
CycleGAN	0.2739	0.5840	13.6223	0.5616
CUT	0.3259	0.6127	12.2771	0.5788
UNSB	0.1304	0.6424	17.5188	0.5255
QS	0.2067	0.8157	12.4637	0.4951
MUNIT	0.0929	0.5430	15.8140	**0.3726**
NOT	0.2971	**1.6266**	31.5810	0.6245
GcGAN	0.1581	0.6030	16.9053	0.5699
CycleDiff	0.2245	0.5998	20.8552	0.5242
Ours	**0.0816**	0.6565	**11.7065**	0.4905

enforces geometric consistency through predefined geometric transformation constraints. NOT [30] employs neural optimal transport to model nonlinear mappings, preserving fine structural details. UNSB [34] is a method based on a neural Schrödinger bridge and stochastic differential equations, which can be regarded as a generalized framework of diffusion models. It proposes a semantically guided strategy specifically designed to address the modality gap in UAV-based IR2Vis translation. In addition, we compare with CycleDiff [44], an SOTA joint learning framework that integrates cycle-consistent translation directly into each iterative denoising step of the diffusion process, thereby optimizing global generative quality.

All baseline methods and the proposed method are trained and tested on Dataset A and Dataset B, with downstream segmentation tasks conducted on Dataset C for further validation. For the translation quality evaluation, we adopt KID, MI, and BRISQUE to assess the translated results from two perspectives: feature distribution alignment and perceptual quality. For the downstream semantic segmentation task, we use mean intersection over union (mIoU), mean pixel accuracy (mPA), and pixel accuracy (PA) to quantitatively evaluate the semantic consistency and practical utility of the generated visible images. All comparative models adopt their official open-source implementations and recommended hyperparameters to ensure rigorous and fair experimental comparisons.

1) Results on Dataset A: Dataset A represents the fundamental challenge of bridging the IR-visible modality gap under consistent daytime illumination. In all quantitative tables throughout this article, the best results are highlighted in bold, and the second-best results are underlined. As shown in Table I, IVT-GSCNet achieves the optimal performance across all metrics on Dataset A, including KID (0.0542), MI (1.3660), and LPIPS (0.6308). Our model outperforms the second-best baseline CUT by 13.8% in KID, demonstrating a stronger capability in fitting the real visible light distribution. Notably, the newly added diffusion-based baseline, CycleDiff, achieves the second-best performance in MI (1.0696), reflecting its capability in maintaining structural layout; however, its visual quality and perceptual fidelity (LPIPS of 0.7085) still lag behind ours.

These quantitative gains are vividly reflected in the qualitative results presented in Fig. 7. From a global perspective, methods such as NOT and GcGAN exhibit severe chromatic instability, frequently producing unnatural green road surfaces and erratic color patches in shaded regions. This suggests that without adaptive noise filtering, these models are prone to overfitting IR thermal intensity fluctuations. Regarding local structural details, the zoomed-in view patches of small vehicles reveal that QS and MUNIT suffer from a "content-loss" effect due to excessive over-smoothing, causing the geometric boundaries of small targets to be obscured by the background. Similarly, although the advanced diffusion-based CycleDiff can restore plausible semantic layouts, its translated micro-scale targets (e.g., the zoomed-in view vehicles) exhibit notable blurriness and edge degradation, failing to reconstruct sharp object boundaries. In contrast, IVT-GSCNet generates clear, noise-free tarmac textures while preserving sharp boundaries for micro-scale targets. This superiority is primarily attributed to the GFRM, which purifies the feature flow at the skip-connection interfaces, effectively shielding the decoder from raw sensor noise propagation.

2) Results on Dataset B: Dataset B poses a significant challenge due to severe temporal and illumination shifts. As summarized in Table II, IVT-GSCNet yields favorable performance in terms of perceptual quality, achieving a KID of 0.0816, a BRISQUE score of 11.7065, and a highly competitive LPIPS of 0.4905. Although the diffusion-based baseline CycleDiff [44] obtains a KID of 0.2245 and an LPIPS of 0.5242, its capability remains limited under such extreme night-to-day domain gaps. While QS yields a higher MI value of 0.8157, the visual evidence in Fig. 8 reveals that its results suffer from a grayscale-ghosting artifact, representing a rigid intensity mapping that fails to reconstruct the rich textures characteristic of daytime visible scenes. Most notably, as shown in the middle row of Fig. 8, CycleGAN suffers from severe structural degradation, where the rigid bridge spans are disintegrated into disorganized, hazy textures. This failure underscores the limitation of cycle-consistency in handling extreme modality shifts. Furthermore, CUT and GcGAN produce prominent checkerboard artifacts in dense vegetation areas. Similarly, the qualitative results of CycleDiff in Fig. 8 exhibit noticeable blurriness, where the vehicle on the bridge and the dense foliage appear washed out and lose fine-grained edge details.

To address the alignment-realism dilemma, IVT-GSCNet relaxes the rigid pixel-wise MI constraints in favor of deep semantic anchoring. This allows our model to generate realistic textures, such as nuanced foliage and shadows, while strictly maintaining the geometric integrity of large-scale rigid structures.

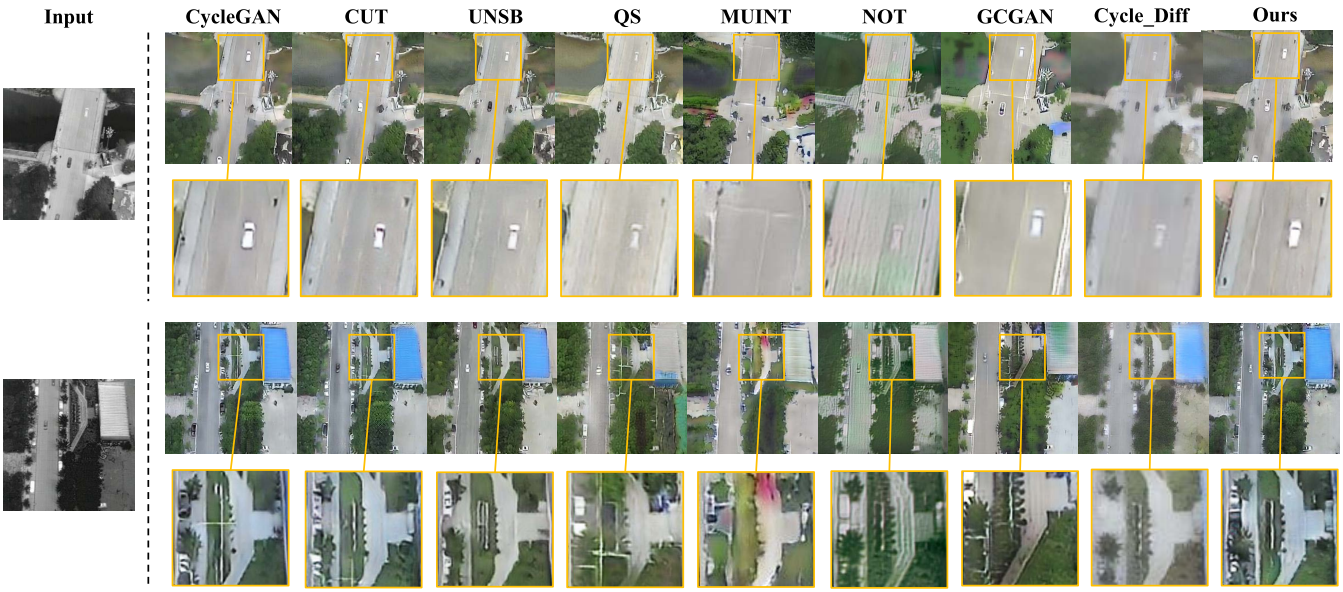


Fig. 7. Qualitative comparison on Dataset A. The left column shows the input IR images, and the right columns show the generated visible images. Yellow highlighted boxes mark the zoomed-in view regions for better detail observation.

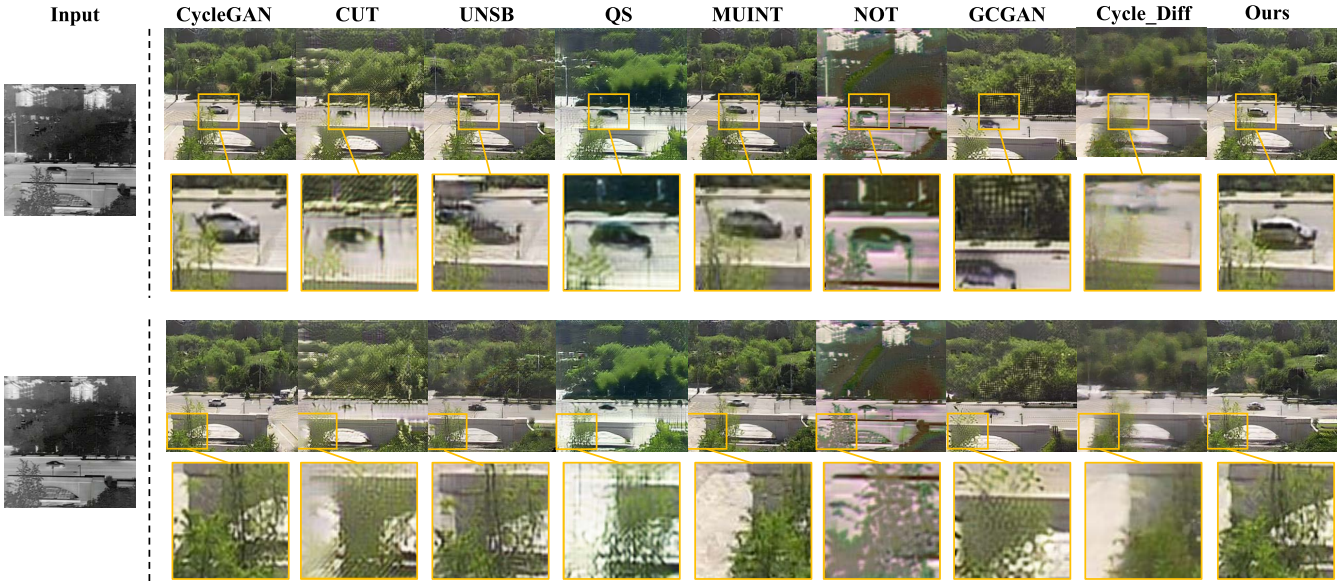


Fig. 8. Qualitative comparison on Dataset B. The left column shows the input IR images, and the right columns show the generated visible images. Yellow highlighted boxes mark the zoomed-in view regions for better detail observation.

3) Results on Dataset C: Dataset C evaluates translation performance across diverse outdoor environments with complex building structures and road targets. As summarized in Table III, IVT-GSCNet achieves favorable results in terms of generative realism and structural preservation, yielding a KID of 0.0876, a BRISQUE score of 10.3061, and an LPIPS of 0.5833. While CycleGAN obtains a higher MI value of 1.3435, its translated outputs exhibit noticeable chromatic distortions and high-frequency noise. In contrast, our model effectively mitigates the alignment-realism tradeoff, generating natural visible textures while maintaining geometric consistency.

These quantitative results are corroborated by the qualitative comparisons in Fig. 9. Globally, GAN-based methods like GcGAN and CycleGAN exhibit severe color artifacts. In the zoomed-in view patches of building windows in the top row, UNSB, CUT, and the diffusion-based CycleDiff suffer from

TABLE III
QUANTITATIVE COMPARISON ON DATASET C

Method	KID(↓)	MI(↑)	BRISQUE(↓)	LPIPS(↓)
CycleGAN	0.0603	**1.6116**	23.0084	0.5469
UNSB	0.0824	0.4972	20.1930	0.5733
QS	0.1255	0.5732	20.3338	0.5891
MUNIT	0.0728	0.4628	27.2303	0.5940
NOT	0.4381	0.3286	23.1720	0.7321
GcGAN	0.0663	0.3967	20.2201	0.5852
CycleDiff	0.1056	0.5629	35.8461	0.6143
Ours	**0.0431**	0.5068	**19.2887**	**0.4758**

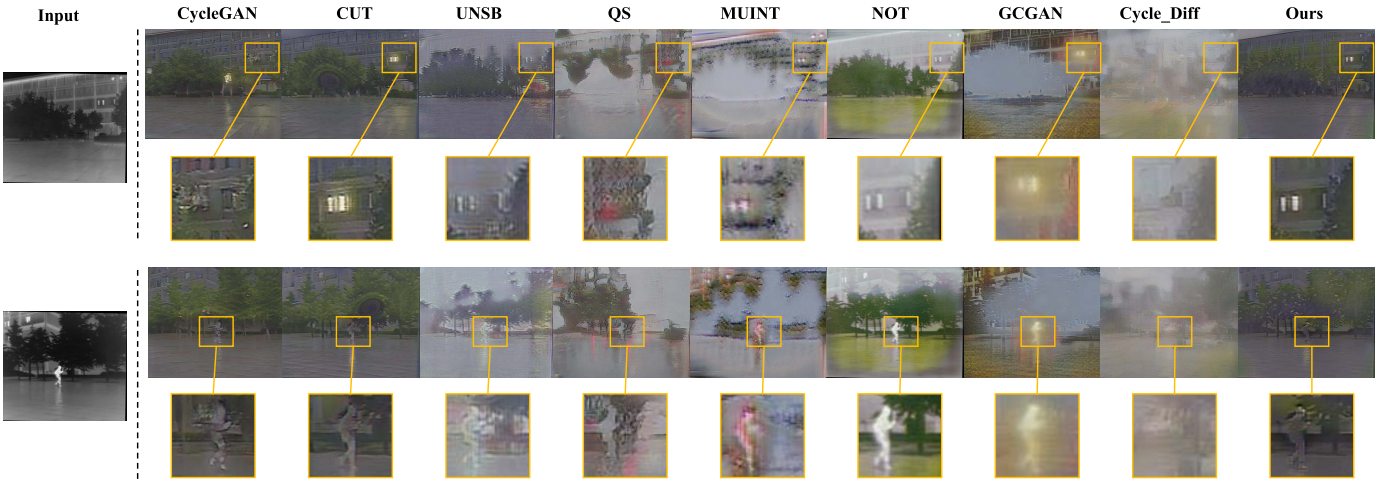


Fig. 9. Qualitative comparison on Dataset C. The leftmost column shows the input outdoor IR images, and the subsequent columns show the generated daytime visible images. Yellow-highlighted boxes mark the zoomed-in view regions for better detail observation.

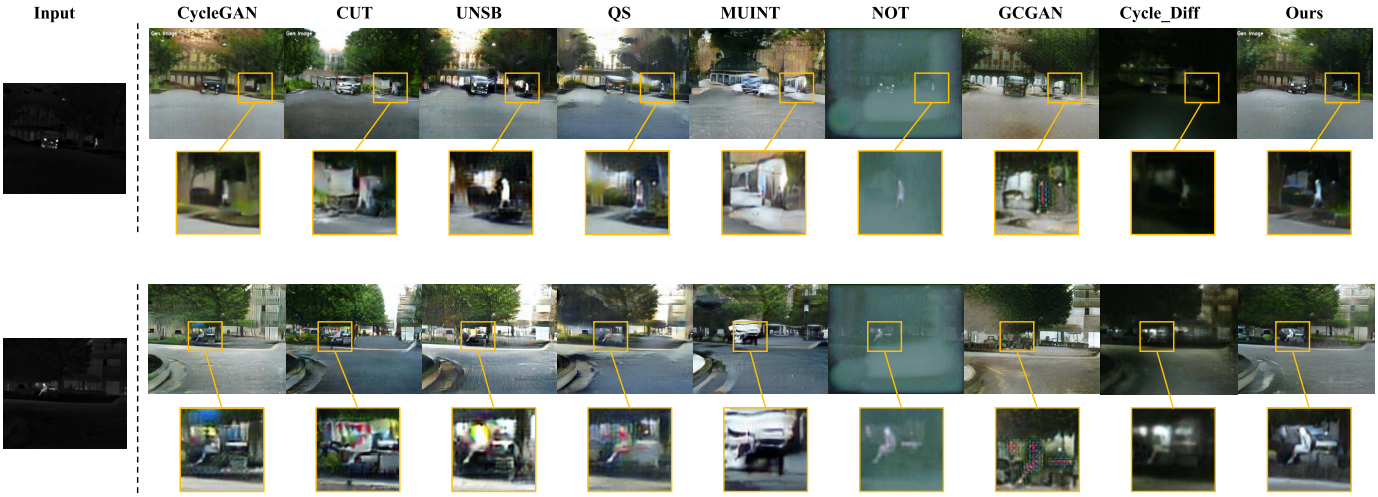


Fig. 10. Qualitative comparison on Dataset D. The leftmost column shows the input nighttime IR images, and the subsequent columns show the generated daytime visible images. Yellow-highlighted boxes mark the zoomed-in view regions for better detail observation.

severe blurring, failing to preserve the distinct window frame structures. Similarly, in the pedestrian walking scenario in the bottom row, while the outputs of CycleDiff appear indistinct, IVT-GSCNet reconstructs a sharp human silhouette with a clear walking posture. This performance is primarily attributed to the GFRM and composite structural constraints, which effectively anchor the structural skeleton from noisy inputs.

4) Results on Dataset D: Dataset D represents an aligned cross-modal evaluation under nighttime-to-daytime conditions. As summarized in Table IV, IVT-GSCNet consistently outperforms other baseline methods in distribution alignment and perceptual quality, achieving a KID of 0.0431 and an LPIPS of 0.4758. Although CycleGAN yields a higher MI value of 1.6116, its synthesized images suffer from style collapse and color deviations, as visually depicted in Fig. 10. Our framework successfully balances these constraints, yielding a BRISQUE score of 19.2887 while preserving fine-grained structural details.

TABLE IV
QUANTITATIVE COMPARISON ON DATASET D

Method	KID(↓)	MI(↑)	BRISQUE(↓)	LPIPS(↓)
CycleGAN	0.0914	**1.3435**	22.3590	0.5869
CUT	0.1302	0.6578	13.0213	0.6583
UNSB	0.2525	0.7698	12.9707	0.6517
QS	0.1470	0.6858	15.9896	0.6747
MUNIT	0.2002	0.8024	16.9109	0.6794
NOT	0.2531	0.7539	30.4822	0.6262
GcGAN	0.1196	0.6753	14.8272	0.6443
CycleDiff	0.1627	0.6849	19.0901	0.6811
Ours	**0.0876**	0.7822	**10.3061**	**0.5833**

Qualitative comparisons in Fig. 10 further support these findings. From a global perspective, GAN-based methods produce unnatural color patches across road surfaces. The diffusion-based baseline CycleDiff reconstructs the basic semantic layout but exhibits noticeable blurriness. In the zoomed-in view patches, the road boundaries generated by

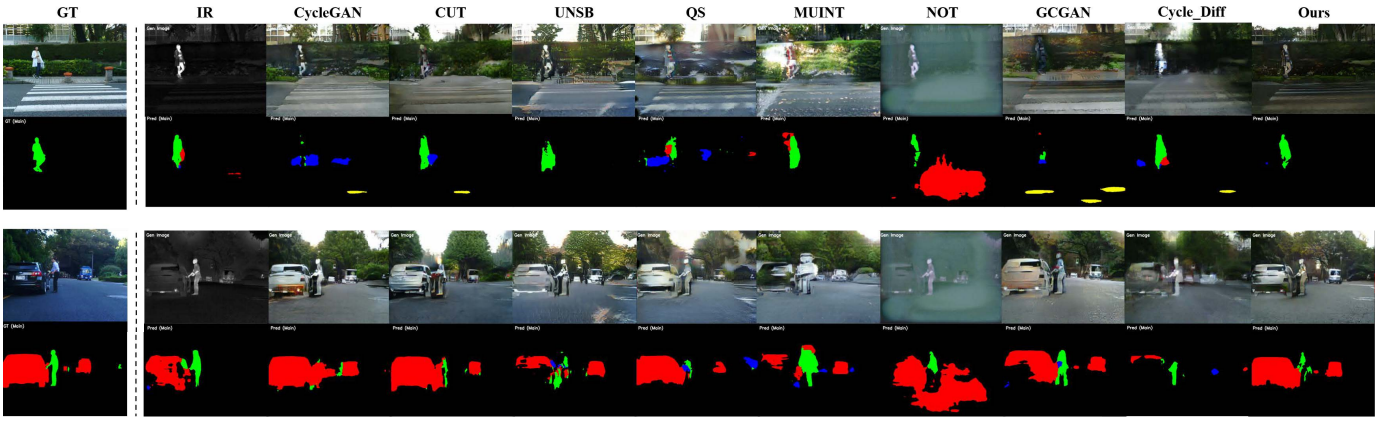


Fig. 11. Qualitative mask visualizations on the MSRS dataset. The first column presents the GT visible images and style labels, the second column shows the input IR images, and the subsequent columns display the synthesized visible images and corresponding segmentation masks generated by different baseline methods and our proposed IVT-GSCNet.

TABLE V
QUANTITATIVE COMPARISON ON DOWNSTREAM SEMANTIC SEGMENTATION PERFORMANCE ON DATASET D

Method	mIoU (%)	mPA (%)	PA (%)	Person IoU	Car IoU	Bike IoU
IR (Baseline)	36.98	42.91	92.97	**43.48**	22.11	7.98
CycleGAN	31.34	37.80	91.80	19.74	25.60	7.56
CUT	19.07	21.54	87.86	1.86	3.93	1.29
UNSB	26.41	32.86	90.56	9.40	19.92	7.17
QS	31.58	36.40	92.44	14.97	23.01	10.17
MUNIT	23.64	27.71	90.81	8.27	14.93	1.71
NOT	23.06	28.06	80.52	28.64	5.12	0.77
GcGAN	29.34	35.32	91.49	13.40	22.75	8.42
CycleDiff	21.42	23.30	91.95	6.11	5.93	2.99
Ours	**37.33**	**43.28**	**93.41**	30.64	**34.29**	**13.67**

TABLE VI
ABLATION STUDY OF PROPOSED MODULES ON THE DATASET A

Method	KID (↓)	MI (↑)	BRISQUE (↓)	LPIPS (↓)
Baseline (CUT)	0.0629	0.9498	20.1492	0.6389
w/o GFRM	0.1266	1.1850	27.2601	0.7069
w/o W-NCE	0.0890	1.0994	24.4457	0.6544
w/o VGG	0.1320	**2.9558**	22.8647	0.7069
w/o Struct	0.0867	0.9590	19.6106	0.6490
Base + GFRM	0.0844	1.0049	19.6825	0.6487
B + G + S	0.2057	2.2015	29.8389	0.8236
B + G + S + V	0.0945	2.2717	22.1077	0.6977
Ours	**0.0542**	1.3660	**18.3021**	**0.6208**

TABLE VII
PARAMETER ANALYSIS OF VARIOUS LAYER-WEIGHTING STRATEGIES ON DATASET A

Method	KID(↓)	MI(↑)	BRISQUE(↓)	LPIPS(↓)
Uniform	0.0641	1.5357	20.2215	0.6530
Shallow-biased	0.0666	**1.6264**	20.3772	0.6493
Deeper-biased	0.0788	1.4267	**17.5068**	0.6708
Extreme Shallow	0.0668	1.6264	20.3772	0.6567
Reverse	0.0705	1.4950	18.6822	0.6591
Ours	**0.0542**	1.3660	18.3021	**0.6208**

CycleDiff appear overly smoothed. In contrast, IVT-GSCNet generates sharp tarmac textures and well-defined pedestrian boundaries, demonstrating that our spatial-channel gating and contrastive strategies can successfully purify thermal features even under extreme low-light constraints.

E. Results on Downstream Semantic Segmentation

Beyond translation quality, we evaluate the utility of the translated images for high-level semantic understanding on Dataset D. We analyze the performance of all methods using a pretrained DeepLabV3+ model; the quantitative results are summarized in Table V, and qualitative mask visualizations are provided in Fig. 11.

As observed in Table V, IVT-GSCNet achieves a top-tier mIoU of 37.33% and a PA of 93.41%. Crucially, most unsupervised baseline methods, such as CUT, UNSB, and NOT, result in performance degradation compared to the raw IR input of 36.98%. This indicates that conventional translation models often introduce structural artifacts that confuse downstream networks. In contrast, our framework achieves a consistent performance gain over the original IR signal, demonstrating its ability to preserve discriminative semantic information. This

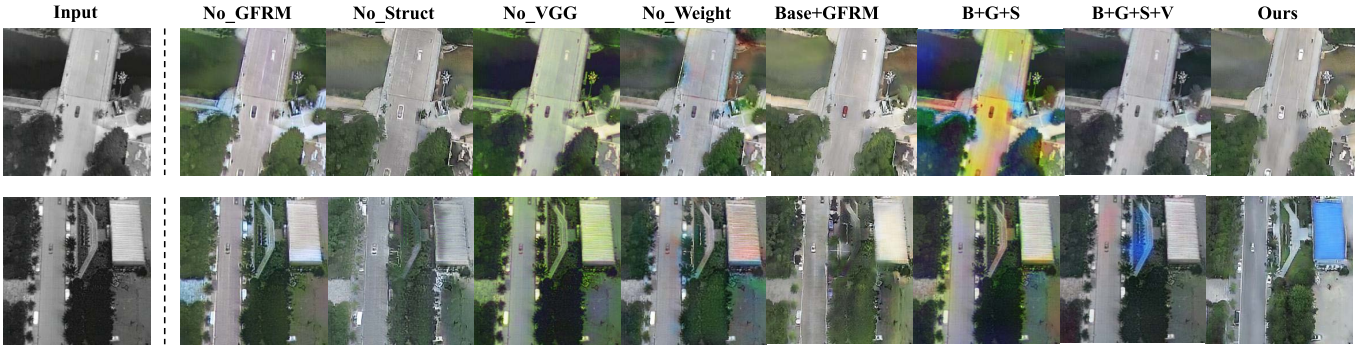


Fig. 12. Qualitative ablation study on Dataset A. (Left to right) Input IR image, w/o GFRM (No_GFRM), w/o Struct (No_Struct), w/o VGG (No_VGG), w/o W-NCE (No_Weight), Base+GFRM, B+G+S, B+G+S+V, and our full model (ours). Notice the severe chromatic hallucinations in B+G+S and B+G+S+V, demonstrating that structural constraints must be paired with layer-weighted contrastive alignment.

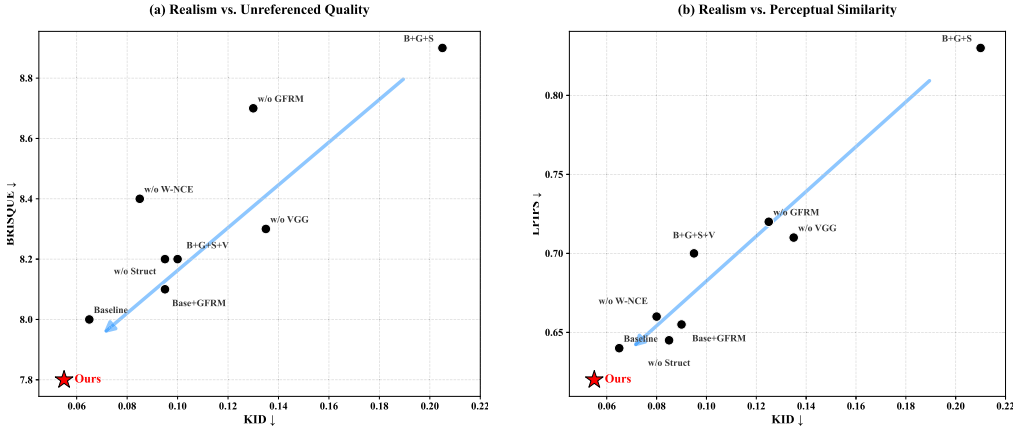


Fig. 13. Scatter plot visualization of the ablation study on Dataset A. (Left) (a) Tradeoff between KID and BRISQUE (scaled for visualization), while (right) (b) shows the tradeoff between KID and LPIPS. The red star denotes our full IVT-GSCNet model, and the blue arrow/trend line highlights the significant optimization trajectory toward the optimal bottom-left "sweet spot."

improvement is particularly pronounced in thermal-sensitive categories, where the IoU for the Car and Bike categories reaches 34.29% and 13.67%, respectively.

The qualitative segmentation maps in Fig. 11 further corroborate these findings. In pedestrian detection scenes in the top row, baselines like MUNIT and NOT produce fragmented masks. In vehicle-heavy scenarios in the bottom row, while CycleGAN and CUT masks appear cluttered, IVT-GSCNet outputs complete masks that accurately align with the ground truth (GT). Ultimately, these results confirm that our model effectively bridges the gap between thermal signals and high-level semantic perception for UAV sensing.

F. Ablation Study

We conduct an extensive ablation study on Dataset A to investigate the individual and synergistic contributions of each component within the IVT-GSCNet framework. Taking CUT as our baseline, we evaluate several model variants including single-module removals like "w/o GFRM," "w/o W-NCE," "w/o VGG," and "w/o Struct," alongside progressive additive combinations such as "Base+GFRM" (where GFRM denotes the gated feature refinement module), "B+G+S" (comprising baseline CUT, GFRM, and composite structural constraints denoted as "S"), and "B+G+S+V" (which further integrates the perceptual color guidance loss denoted as "V"). The quantitative results are summarized in Table VI while the performance tradeoffs are visualized in Fig. 13, and qualitative comparisons are shown in Fig. 12.

1) Quality and Realism Analysis: The visualization in Fig. 13(a) demonstrates the tradeoff between image realism and unreferenced quality. The results show that the ablation of the GFRM leads to the most substantial performance drop. Specifically, the BRISQUE score rises to 27.2601 and the KID increases to 0.1266 when GFRM is removed. This confirms that GFRM is essential for filtering IR sensor noise and preventing signal contamination in the decoder. This statistical degradation is vividly reflected in Fig. 12, where the "No GFRM" model exhibits granular artifacts in tree regions, whereas the GFRM in our full model serves as an indispensable gatekeeper for feature purity. The blue optimization trajectory in Fig. 13 illustrates how our full model effectively pulls the baseline performance toward the optimal bottom-left region, achieving a KID of 0.0542 and a BRISQUE score of 18.3021, which marks the highest realism and quality among all tested variants.

2) Perceptual Similarity and Realism Analysis: The relationship between image realism and perceptual similarity is further illustrated in Fig. 13(b). The removal of our layer-weighted contrastive strategy (w/o W-NCE) results in a noticeable drift in both dimensions where the KID reaches

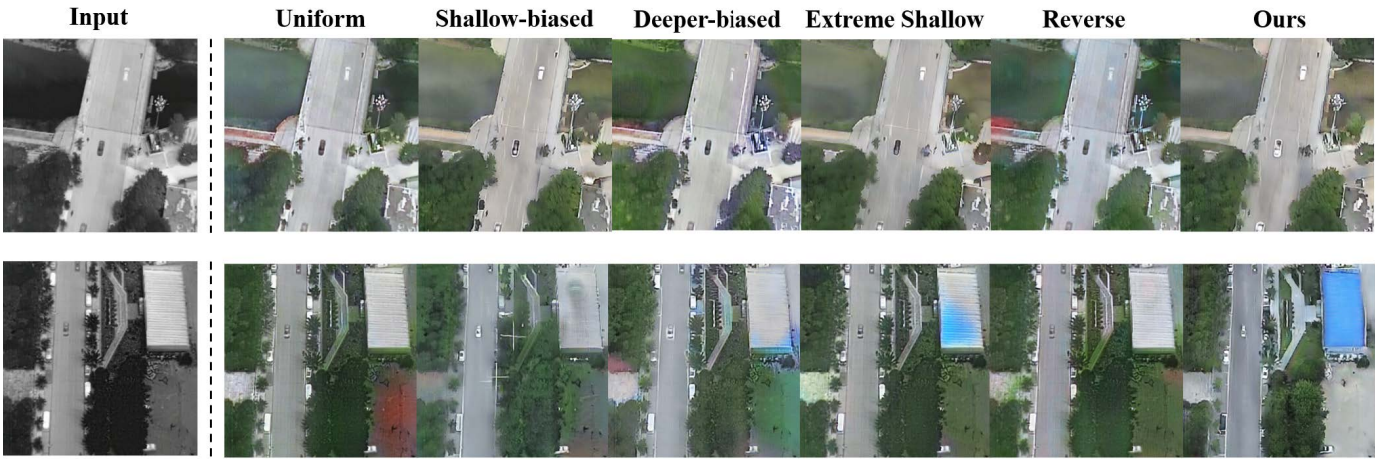


Fig. 14. Qualitative comparison of different weighting strategies on Dataset A. (Left to right) Input IR image, uniform, shallow-biased, deeper-biased, extreme shallow, reverse, and ours.

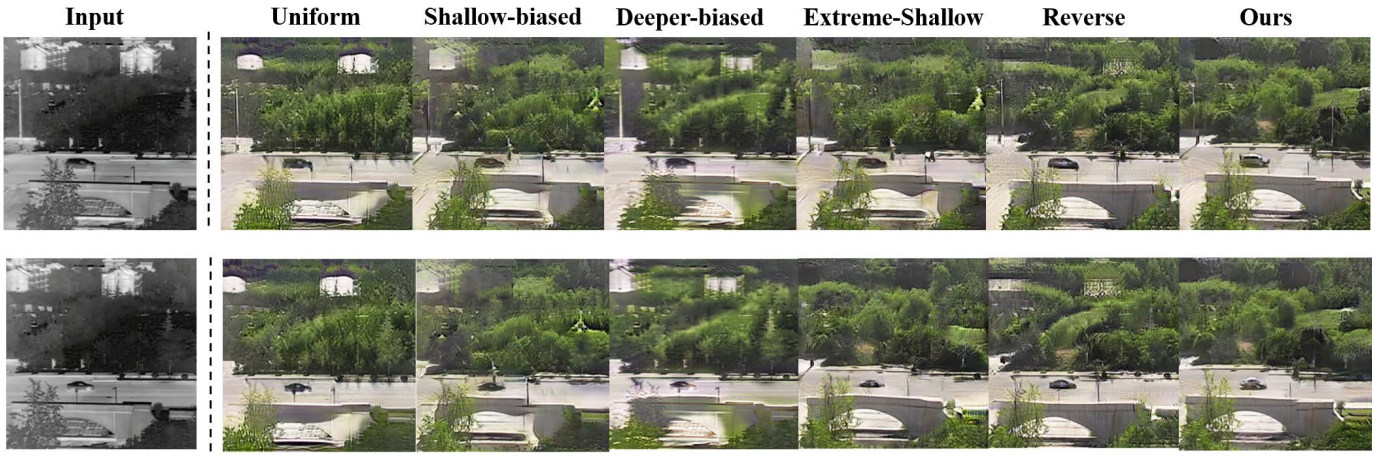


Fig. 15. Qualitative comparison of different weighting strategies on Dataset B. (Left to right) Input IR image, uniform, shallow-biased, deeper-biased, extreme shallow, reverse, and ours.

0.0890 and the LPIPS degrades to 0.6540. This outcome underscores the effectiveness of our hierarchical alignment approach in anchoring deep semantic logic. Furthermore, we observe that the "B+G+S" variant produces the poorest perceptual fidelity with an LPIPS of 0.8236. This suggests that imposing rigid structural constraints without the coordination of perceptual color guidance can over-regularize the generator and lead to unnatural textures. In contrast, the full IVT-GSCNet harmonizes these constraints to achieve the best LPIPS of 0.6208. These joint tradeoffs in Fig. 13 prove that our proposed modules function as a highly coordinated system rather than a collection of independent additive components, reaching the sweet spot of performance represented by the red star.

G. Parameter Analysis

To investigate the sensitivity of the hierarchical layer-weighting strategies, we evaluate six configurations across the encoder hierarchy, ranging from focus on low-level textures to high-level semantics. Quantitative results are summarized in Tables VII and VIII, while corresponding qualitative comparisons and performance evolutionary trends are visualized in Figs. 14–16, respectively.

TABLE VIII
PARAMETER ANALYSIS OF VARIOUS LAYER-WEIGHTING STRATEGIES ON DATASET B

Method	KID(↓)	MI(↑)	BRISQUE(↓)	LPIPS(↓)
Uniform	0.1562	0.7927	14.9925	0.5484
Shallow-biased	0.1771	<u>0.8595</u>	14.8107	<u>0.5209</u>
Deeper-biased	0.2281	**0.8977**	24.4635	0.5468
Extreme Shallow	<u>0.1083</u>	0.6721	13.8968	0.5365
Reverse	0.1553	0.7020	<u>12.9151</u>	0.5424
Ours	**0.0816**	0.6565	**11.7065**	**0.4905**

1) Evaluation on Dataset A: Dataset A assesses stylistic consistency under stable illumination. As illustrated by the V-shaped optimization trajectories in Fig. 16, our proposed strategy consistently achieves the global minimum for both KID (0.0542) and BRISQUE (18.30). Crucially, our balanced strategy also secures the best perceptual similarity with the lowest LPIPS score of 0.6208 in Table VII, demonstrating its superior capability in reconstructing human-aligned visual features.

A notable observation in Table VII is the MI; the *extreme shallow* variant yields a higher MI (1.6264) than

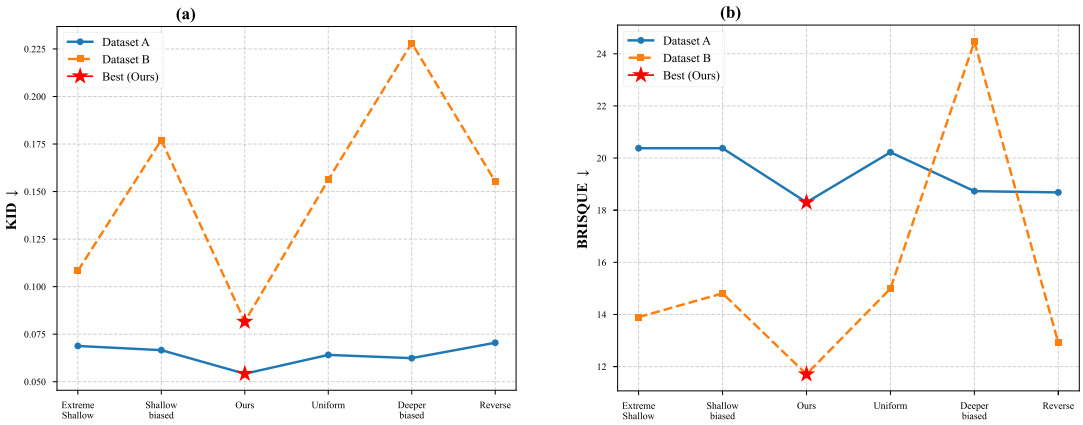


Fig. 16. Sensitivity analysis of the hierarchical layer-weighting strategies across six strategic configurations. The red star denotes our proposed balanced scheme. (a) Distribution Fidelity (KID). (b) Perceptual quality (BRISQUE).

our method, yet its visual realism is significantly deficient. The qualitative results in Fig. 14 reveal the underlying cause: over-emphasizing shallow weights forces a rigid, deterministic intensity mapping. This maintains strict pixel-wise correlation at the expense of stochastic textural realism, resulting in a "colored-grayscale template" appearance. In contrast, our balanced weighting scheme appropriately relaxes these rigid constraints, granting the model generative flexibility to synthesize vibrant textures while anchoring on the global semantic manifold.

2) Evaluation on Dataset B: Dataset B poses a more significant challenge involving extreme temporal and illumination shifts, where the role of spatial anchors becomes paramount. A precipitous degradation is observed for the *deeper-biased* strategy, where the metrics in Fig. 16 exhibit a sharp spike. This quantitative failure is visually corroborated in Fig. 15, where bridge spans and tree textures appear hazy and indistinct. This confirms that neglecting shallow-layer weights causes the model to lose fine-grained spatial gradients—the "geometric skeleton" of the scene—leading to a total collapse of structural clarity.

Our method achieves the optimal KID (0.0816), BRISQUE (11.71), and LPIPS (0.4905) in Table VIII by navigating the inherent alignment-realism dilemma. By assigning lower but nonzero weights to shallow layers and higher weights to deep layers, our strategy effectively utilizes shallow features as spatial anchors to "fix" pixels to their precise coordinates without over-constraining the synthesis of daytime-consistent textures (e.g., nuanced shadows). This equilibrium ensures that the generated images are both structurally coherent with the nighttime source and perceptually indistinguishable from real daytime photographs.

3) Summary of Synergy and the Evaluation Paradox: The ablation experiments and the optimization path highlighted by the blue arrow in Fig. 13 demonstrate that IVT-GSCNet is not a mere collection of additive parts but a holistically integrated architecture. The synergistic relationship between the modules is critical where the GFRM acts as a prerequisite denoising gateway that ensures the layer-weighted contrastive learning aligns meaningful semantic structures rather than sensor noise. Subsequently, the composite structural constraints provide a geometric anchor that allows the perceptual guidance to synthesize realistic textures without causing boundary drift or structural deformation.

Notably, the results reveal an evaluation paradox regarding individual metrics, particularly the MI. As observed in Table VI, the variant without VGG yields a paradoxically high MI score of 2.9558 which numerically suggests superior alignment. However, this variant actually suffers from catastrophic mode collapse and produces illogical color patches that fail to represent realistic visible-light scenes. This proves that a singular pursuit of high MI can lead to a correlation trap where the model overfits to a deterministic but nonphysical mapping.

In contrast, our full model intentionally navigates the alignment-realism dilemma and resides at the performance sweet spot represented by the red star in both the quality analysis of Fig. 13 and the similarity analysis of Fig. 13. This position represents the optimal balance where our framework slightly relaxes rigid pixel-wise correlation to grant the generator the flexibility required to synthesize photorealistic textures. Collectively, this synergy addresses the triad of challenges in UAV sensing comprising noise, misalignment, and modality distortion, resulting in an output that is both structurally consistent and perceptually authentic.

H. Parameter Analysis

To investigate the sensitivity of the hierarchical layer-weighting strategies, we evaluate six configurations across the encoder hierarchy where the focus ranges from low-level textures to high-level semantics. The quantitative results for Dataset A and Dataset B are summarized in Tables VII and VIII, respectively, while the corresponding performance evolutionary trends are visualized in Fig. 16.

1) Evaluation on Dataset A: Dataset A assesses stylistic consistency under stable illumination conditions. As illustrated by the optimization trajectories in Fig. 16, our proposed weighting strategy reaches the global minimum for both image realism and perceptual similarity. Specifically, our method achieves the best KID of 0.0542 and the lowest LPIPS score of 0.6208, as shown in Table VII. Although the deeper-biased variant yields a slightly better BRISQUE score of 17.5068, its higher KID indicates a loss of generative realism compared to our full model.

A notable observation in the parameter analysis is the behavior of the MI metric. The Shallow-biased and extreme shallow variants yield the highest MI score of 1.6264 which numerically suggests a tighter alignment with the source. However, the qualitative results reveal that over-emphasizing shallow weights forces a rigid and deterministic intensity mapping. This maintains strict pixel-wise correlation at the expense of stochastic textural realism and results in a colored-grayscale appearance. In contrast, our balanced weighting scheme appropriately relaxes these rigid constraints to grant the model the generative flexibility required to synthesize vibrant textures while anchoring on the global semantic manifold.

2) Evaluation on Dataset B: Dataset B poses a more significant challenge involving extreme temporal and illumination shifts where the role of spatial anchors becomes paramount. We observe a precipitous degradation for the deeper-biased strategy as the metrics in Fig. 16 exhibit a sharp spike. According to Table VIII, the KID for this variant rises to 0.2281 and the BRISQUE score reaches 24.4635. This quantitative failure suggests that neglecting shallow-layer weights causes the model to lose fine-grained spatial gradients which represent the geometric skeleton of the scene and leads to a collapse of structural clarity.

Our balanced strategy successfully navigates this alignment-realism dilemma and achieves the optimal scores across nearly all key metrics, including a KID of 0.0816, a BRISQUE of 11.7065, and an LPIPS of 0.4905. By assigning lower but nonzero weights to shallow layers and higher weights to deep layers, our strategy effectively utilizes shallow features as spatial anchors to fix pixels to their precise coordinates without over-constraining the synthesis of daytime-consistent textures. Although the deeper-biased variant achieves a higher MI of 0.8977, our method ensures that the generated images are both structurally coherent with the nighttime source and perceptually indistinguishable from real daytime photographs, marking the best overall performance.

IV. Conclusion

This article proposes IVT-GSCNet, a robust framework designed specifically for high-fidelity IR2Vis image translation. To address the three core challenges of noise propagation, semantic misalignment, and geometric distortion, the framework integrates three synergistic innovations: an end-to-end encoder–decoder architecture that maintains global semantic consistency and local details when processing high-noise inputs, a GFRM in the skip connections that suppresses noise and enhances small-target features through a spatially adaptive channel attention mechanism, and a layer-weighted contrastive learning and composite structural constraint mechanism that utilizes multilayer weighted PatchNCE combined with edge loss, structural similarity constraints, and TV constraints to achieve spatial-semantic alignment and geometric structure preservation.

Extensive quantitative and qualitative evaluations on two challenging cross-modal datasets demonstrate that IVT-GSCNet significantly outperforms current SOTA methods, particularly in distribution fidelity and perceptual quality. Notably, our parametric analysis reveals a fundamental alignment-realism dilemma in cross-modal translation. By employing a strategically balanced layer-weighting scheme, our model achieves a superior tradeoff that prioritizes generative photorealism while maintaining rigorous structural coherence. This work not only provides a practical solution for IR scene reconstruction in UAV sensing but also advances the theoretical understanding of hierarchical feature alignment in contrastive learning.

While IVT-GSCNet yields favorable translation performance, several limitations remain to be addressed in future research. First, the model's robustness under extreme weather conditions such as heavy rain and dense fog, or under drastic illumination changes, requires further validation. Second, although the GFRM is designed to be lightweight, it still introduces minor computational overhead, necessitating further model compression and optimization before deployment on ultralow-power, resource-constrained UAV edge hardware. Finally, the reconstruction fidelity for extremely small targets of less than 10 pixels under complex background clutter still has room for improvement. Future work will focus on temporal consistency modeling, model optimization for edge devices, and small-target preservation.

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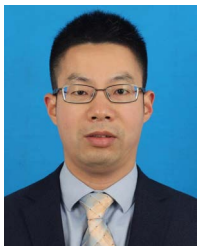
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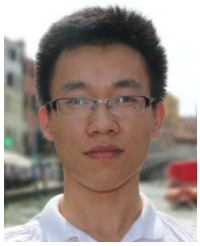
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